

1. Configuration of Network Devices using Packet Tracer tools (Hub, Switch, Ethernet, Broadcast).

Experiment - 1

AIM: To study the following network devices in detail

- * PC
- * Server
- * Repeater
- * Hub
- * Switch
- * Bridge
- * Router
- * Gateway
- * Transmission medium

Apparatus (software): Cisco Packet Tracer

1. Node: In a communications network, a network node is a connection point that can receive, create, store or send data along distributed network routes.

2. Repeater :

A repeater is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances.

3. Hub :

Hub or concentrator is a device for connecting multiple twisted pair or fiber optic ethernet devices together and making them act as a single network segment. Hubs work at the physical layer of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

4. Switch :

A network switch or switching hub is a networking device that connects and sends data only to the correct device. It works mainly at the data link layer.

layer 2). Some advanced switches also work at layer 3.

5. Bridge:

A bridge connects two or more network segments and works at layer 2. It checks data and forwards it only to the correct segment, reducing unnecessary traffic. A switch is basically a multi-port bridge.

6. Router:

A router connects two or more networks and sends data between them using the best path. It reads the address in each packet and decides whether to keep it in the same network or send it to another network. Routers are used for routing tables to choose best route.

7. Gateway:

A gateway connects two different networks and translates data between them so they can communicate.

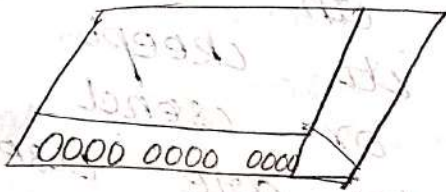
8. Server:

A server is a computer that provides services or resources to other computers on a network.

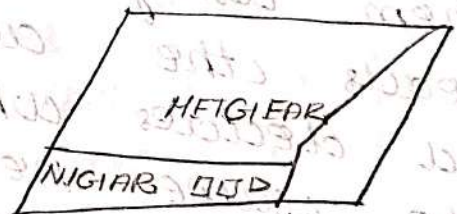
9. Transmission Media:

The paths through which signals travel between devices. They are of two types: guided cables and unguided cables (wireless).

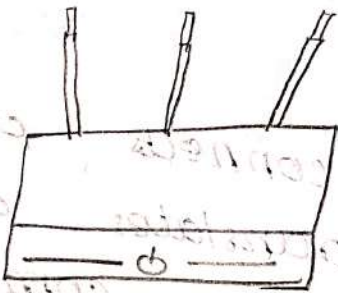
DIAGRAM:



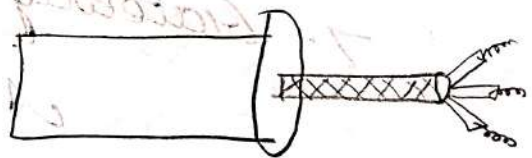
Switch



Bridge



Router



Cable

RESULT:

Thus the result has been executed successfully.

Dr. M. B. M.

Experiment - 2

Implementation of star Topology

Using Tracer.

AIM:

To implement a star topology using packet tracer and hence to transmit data between the devices connected using star topology.

Apparatus Required:

Step 1: Start Packet Tracer

Step 2: choosing devices and communication.

Step 3: Building the technology - adding that single click on the end devices. single click on the Generic hosts area. Move the cursor into topology area and single click in the topology area and it copies.

Step 4: Building the topology - connecting the host to switches. select a switch by clicking once a switch and once on a switch.

Add the switch by moving the plus sign.

Step 5: Connect PCs to switch by first choosing connection. click once on the copper straight through cable click once on PC2 choose Fast Ethernet. Drag the cursor to switch c. click once on switch c).

Step 6: Configuration IP Addresses and subnet masks on the host.

To start communication between the hosts IP addresses and subnet masks, have to be configured on the devices. click once on PC0. choose the config tab and click on Fast Ethernet c. type the IP address in its field. click on the subnet mask it will be generated automatically.

Step 7.

To confirm data through transfer between the devices.

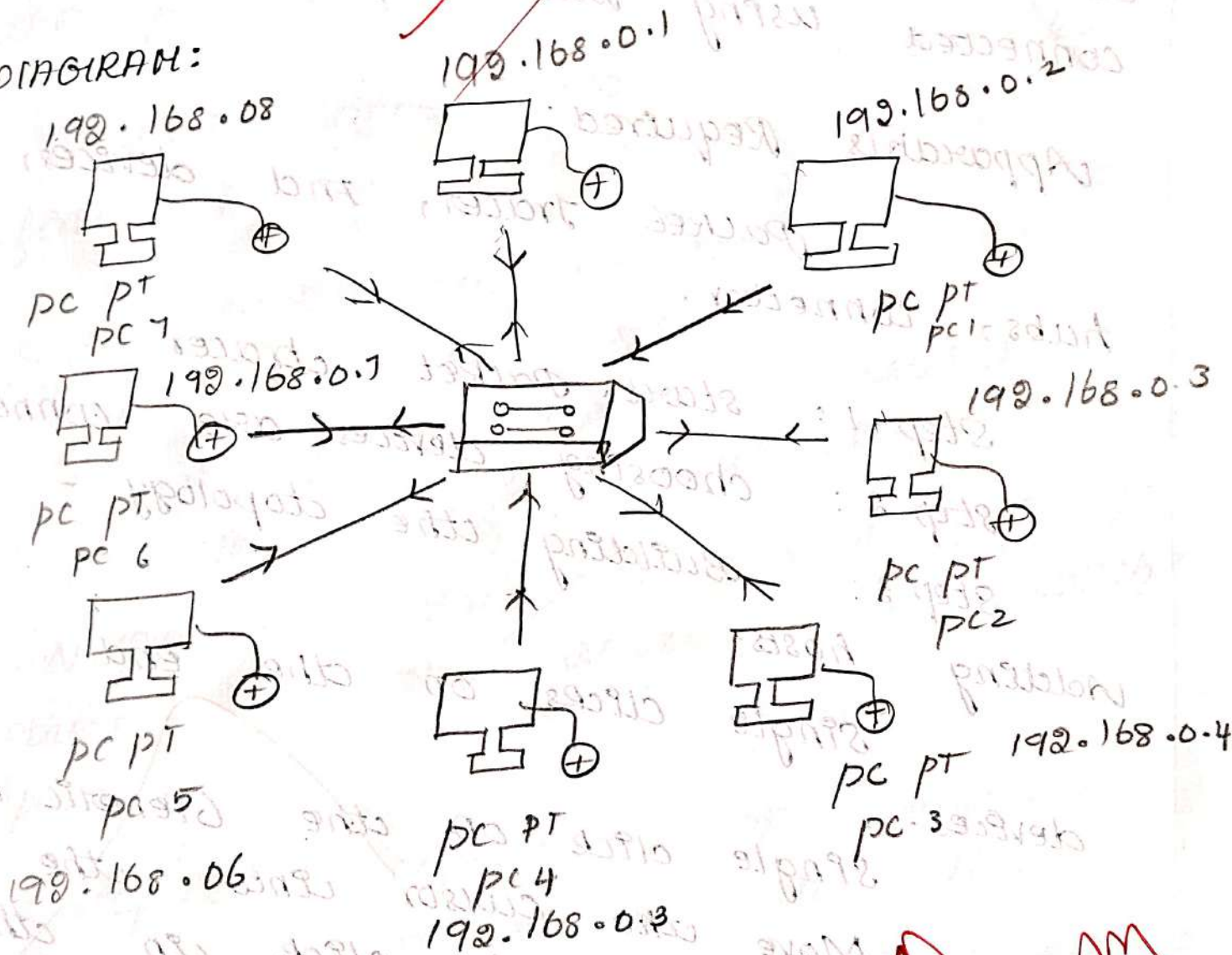
click on the node.

select desktop option and
command prompt

once the window pops up,
ping the IP address of the
device to which node C is
connected.

Ping statistics will be displayed.

DIAGRAM:



RESULT:

Thus the output has
been executed successfully

Drum

Experiment - 3

Implementation of Bus Topology Using Packet Tracer

AIM:

To implement a bus topology using packet tracer and hence to transmit data between the devices connected using Bus topology.

Apparatus Required:

Packet Tracer, End devices, hubs, connector.

Step 1: start packet tracer

Step 2: choosing devices and connections

Step 3: Building the topology -

Adding hosts

Single clicks on the end devices.

Single click on the Generic host

Move the cursor into the

topology area single click on the topology area and it copies the

device.

Step 4: Building the topology - connecting the hosts to switches.

Select a switch by clicking once on switches and once on 2950 - 24 switch.

Add the switch by moving the plus sign +.

Step: 5 Connect PCs to switch by first choosing connections.

Click once on PC2

Choose Fast ethernet

Drag the cursor to switch (1)

Click once on switch (2)

Notice the Green link lights on PC ethernet NIC and amber light switch port.

Step: 6 Configuring IP Addresses and subnet masks on the Hosts.

Open PC → config → Fast ethernet (1)

Ethernet the IP Address
and the subnet masks the Device
will now be ready for communication
Step: 7 To confirm data transfer
between the devices

click on the node
Select desktop option and
command prompt

Once the windows pops up,
ping the IP address of the
devices to which node C is
connected.

Ping statistics will be displayed.

DIAGRAM:

192.168.0.1



PC PT
PC 8



2960-24TT

switch 0

2960-24TT

switch 1



PC PT

PC 9

192.168.0.2



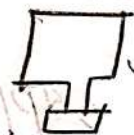
2960-24TT

switch 2



2980-24TT

switch 3



PC PT

PC 11

192.168.0.4

11/11/11

RESULT:

Thus the output has been executed successfully.

Experiment - 4

Implementation of Ring Topology Using Tracer

AIM:

To implement a ring topology using packet tracer and hence to transmit data between the devices connected using Ring topology.

Apparatus Required:

Packet tracer, Hubs, connectors

Steps for building topology: L

Step: 1 Start Packet tracer

Step: 2 choosing devices and connections

Step: 3 Building the topology -

Adding hosts single click on the end devices.

single click on the end devices.

single click on Generic host

Move the cursor into topology area.

Step: 4 Building the topology - connecting

the hosts to cswitches

Select a switch by clicking once on cswitches, and once on a 2950-24 cswitch

Add the switch by moving the plus sign +

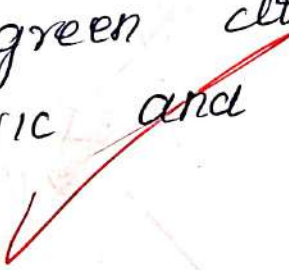
Step 5: Connect pcs to cswitch by first choosing connections

click once on p2

choose Fast ethernet

drag the cursor to cswitch c)

click once on cswitch c)

Notice the green link lights on pc ethernet switch port.  NIC and amber light

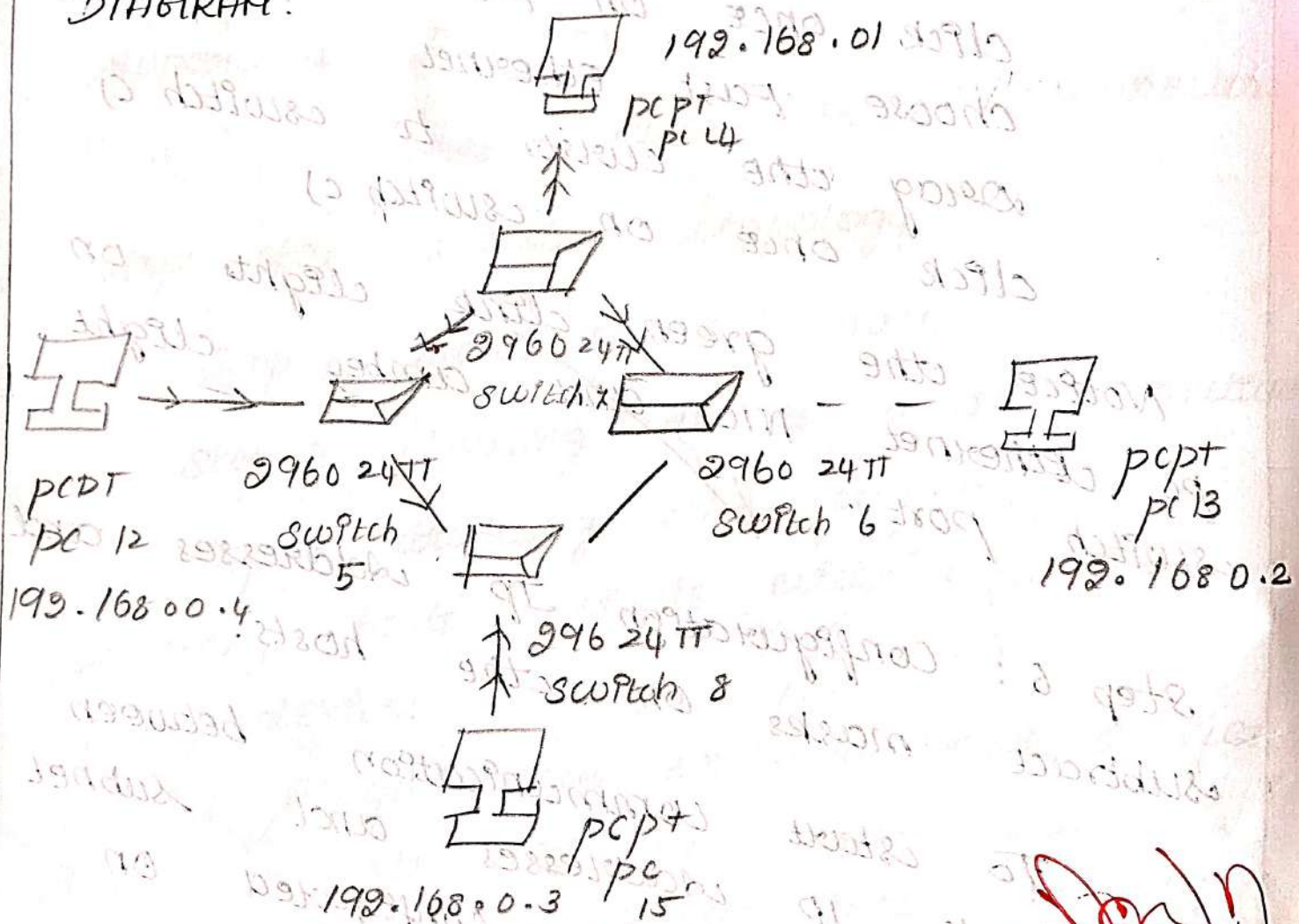
Step 6: Configuration IP Addresses and Subnet masks on the hosts.

To start communication between the hosts IP addresses and subnet masks had to be configured on devices.

Step 7: To confirm data transfer between the devices

click on the node select desktop option then command prompt once the windows pop up ping the IP addresses of the device to which model c) is connected.

DIAGRAM:



RESULT:

Thus the output has been executed successfully.

Experiment - 5

Implementation of Mesh Topology using Packet Tracer

AIM:

To implement a mesh topology using packet tracer and hence to transmit data between the devices connected using mesh topology.

Apparatus Required:

Packet tracer, Hubs, connectors

Steps for building topology:

Step: 1 start Packet tracer

Step: 2 choosing devices and connectors - Add

Step: 3 Building the topology

hosts single click on the end devices
single click on the Generic host
Move the cursor into topology

area
Step: 4 select a switch, by clicking
once on switches and once on a

2950 - 24 cswitch

add the cswitch by moving
the plus sign to

Step 5: Connect pc's to cswitch by

first

click once on pc

choose Fast ethernet

drag the cursor to cswitch c)

click once on cswitch c)

Notice the green link lights on pc
Ethernet NIC and amber light cswitch

Port.

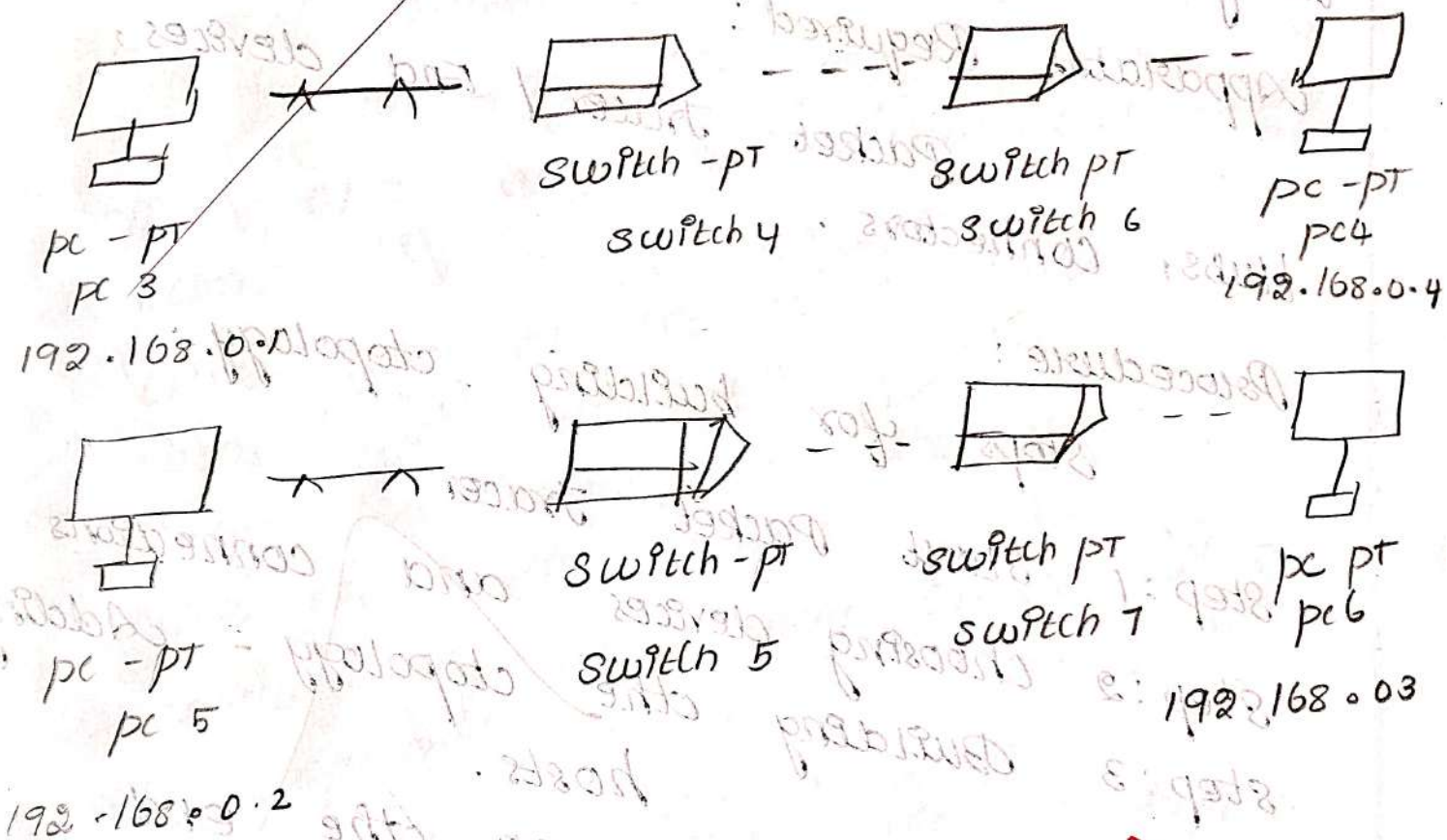
Step 6: Configuring IP Address and
subnet masks on the Hosts

To start communication between the
hosts IP Address and subnet Mask had
to be configured on the devices.
click once on pc. Choose the config
tab and click on fast ethernet 0. Type
the IP address in its field. click
on the subnet mask it will be
generated automatically.

Step 7: To confirm data transfer between the devices.

Click on the node. select desktop option and then command prompt. Once the window pops up, ping the IP address of the device to which node 0 is connected. Ping statistic will be displayed.

DIAGRAM:



RESULT:

Thus the output has been executed successfully.

Experiment - 6

Implementation of Tree topology using Packet Tracer

Aim: To implement a tree topology using packet tracer and hence to transmit data between the devices connected using tree topology.

Apparatus Required:

Packet Tracer / End devices,
Hubs, connectors.

Procedure:

Steps for building topology:

Step: 1 Start Packet Tracer

Step: 2 choosing devices and connections

Step: 3 Building the topology - Adding hosts.

Single click on the end devices

Single click on the Generic host

Move the cursor into topology area.

Single click in the topology

area and it copies the device

Step 4: Building the star Topology -

connecting the hosts to hubs.

select a Hub, by clicking once on

Hub and once on a generic hub.

Add the Hub by moving the plus

sign +

Step 5: Connect PCs to Hub by first

choosing connections

click once on the automatic cable

selector.

click once on PC2

choose Fast ethernet

drag the cursor to Hub

click once the Hub

Proceeding in this way create three

star topologies.

Step 6: Building the tree Topology -

connecting the hubs to active hub

connects the hubs of star

topologies to active hub to create

tree topology.

Step 7: Configuring IP Address and Subnet masks on the hosts.

To start communication between the hosts IP Addresses and Subnet masks had to be configured on the devices: click once on PC0, choose the config tab and click on fast ethernet 0. Type the IP address in its field. Click on the subnet mask. It will be generated automatically.

Step 8: Verifying Connectivity in Real time Mode

Be sure you are in Real time mode. Select the add simple pdu tool used to ping devices. click once on PC0, then once on PC3. The pdu last status should show as successful.

Step 9: Verifying Connectivity in Simulation Mode

Be sure you are in simulation mode.

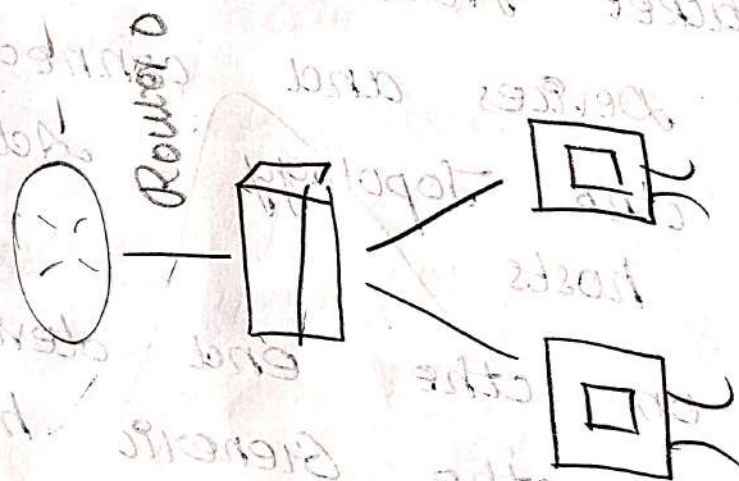
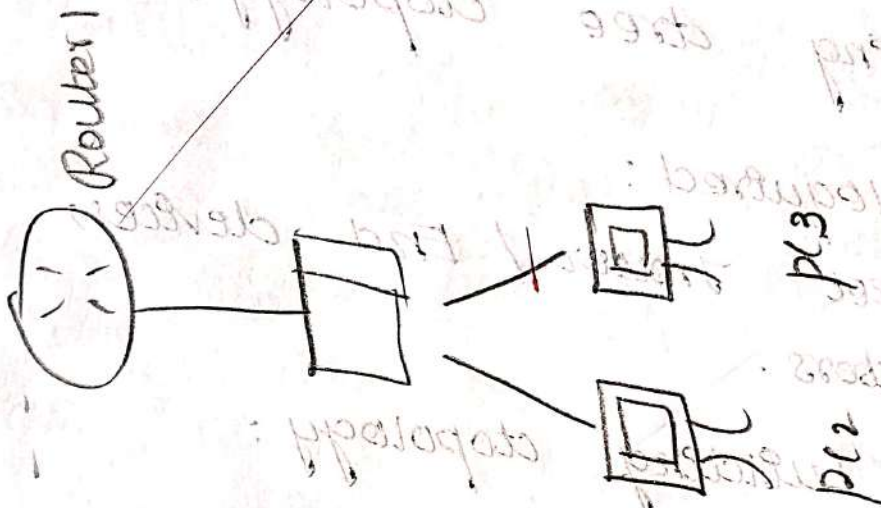
Deselect all filters (All/none) and select only ICMP.

Select the add simple pdu tool

used to ping devices.

click once on PC0, then once on PC3.
Continue clicking capture / forward button
until the ICMP ping is completed.
you should see the ICMP messages
move between the hosts, hub
and switch. The PDU last status
should show as successful.

DIAGRAM:



RESULT:

Thus the program
executed successfully.

WAB

Experiment - 7

Implementation of Hybrid Topology (Bus and Ring Topology) using Packet Tracer.

Aim: To implement a hybrid topology using packet tracer and hence to transmit data between the devices connected using tree topology.

Apparatus required:

Packet Tracer / End devices, Hubs, connectors.

Steps for building topology:

Step 1: Start Packet Tracer

Step 2: choosing devices and connectors

Step 3: Building the topology - Adding hosts

Single click on the end devices

Single click on the Generic host.

Move the cursor into topology area.

Single click in the topology area and it copies the device.

Step 4: Building the Bus topology -
Connecting the hosts to hubs.

Select a hub, by clicking once on
hub and once on a generic hub
Add the hub by moving the
plus sign '+'

Step 5: Building the Ring Topology -
Connecting the hosts to Hubs.

Select a Hub, by clicking once on
hub and once on a generic Hub
Add the hub by moving the plus
sign '+'

Step 5: connect PCs to hub by first
choosing connections
click once on the Automatic cable
selector.

click once on PC2
choose Fast ethernet
Drag the cursor to Hub0
click once on Hub0

Proceeding in this way create
three Bus topologies.

Step 6: Building and connecting the hubs to Active hub.

Connect the hubs of star topologies to active hub to create tree topology.

Step 7: Configuring IP Addresses and subnet masks on the Hosts.

To start communication between the hosts IP Address and subnet masks had to be configured on the devices. click once on PC0 choose the config tab and click on Fast ethernet 0. Type the IP address in its field. click on the subnet mask. It will be generated automatically.

Step 8: Verifying connectivity in

Realtime Mode

Be sure you are in Realtime mode.

Select the add simple PDU tool used to ping devices.

click once on PC0, then once on

PC3

Step 6: Building the tree. Topology.
connecting the hubs to Active hub.

Connect the hubs of star
topologies to active hub to create
tree topology.

Step 7: Configuring IP Addresses and
subnet masks on the Hosts.

To start communication between
the hosts IP Address and subnet
masks had to be configured on the
devices. Click once on PC0 choose the
config tab and click on Fast ethernet 0.
Type the IP address in its field.
Click on the subnet mask. It will
be generated automatically.

Step 8: Verifying connectivity in

Realtime Mode

Be sure you are in Realtime
mode.

Select the add simple PDU tool
used to ping devices.

click once on PC0, then once on

The PDU Last status should show as successful.

Step 9: Verifying connectivity in Simulation Mode

Be sure you are in simulation mode.

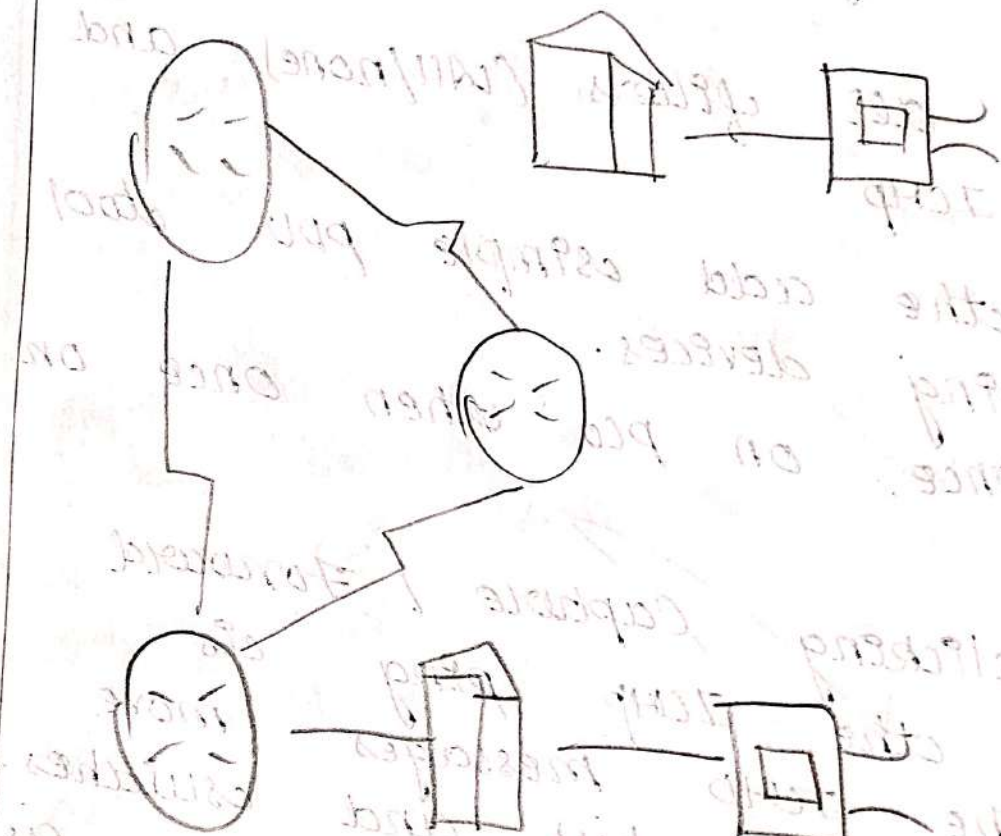
Deselect all filters (All/none) and select only ICMP

used to ping the add sample PDU tool click once on p1 then once on

pc3.

Continue clicking Capture / Forward button until the ICMP ping is completed. The ICMP messages move between the hosts, hub and switches. The PDU Last status should show as successful.

DIAGRAM:



RESULT:

Down

Thus the program is
executed successfully.

Experiment - 8

Data link layer Traffic simulation
Using Packet Tracer Analysis
of ARP.

AIM:

To implement Data link layer
Traffic simulation Using Packet Tracer
Analysis of ARP.

Software Required:

Packet Tracer / End devices,
switches, connectors.

Requirements:

1. End device - They are the
devices through which we can pass
message from one device to another
and they are interconnected.

2. Switch / Hub - Interface between
two devices.

3. Cable - Used to connect two
devices.

Procedure:

1. open Packet Tracer
2. click on the list the available capture interface.
3. choose the pc, server and Hub.
4. Later given connection from hub to the remaining pcs.
5. Give Ip address to the pcs with configuration
6. Simulate the source and destination.

RESULT:

Thus the program has been executed successfully.

W. S. R.

Experiment - 9

Data link layer Traffic simulation
using Packet Tracer Analysis of
CSMA/CD and CSMA/CA.

AIM: To implement data link layer
Traffic simulation using Packet Tracer
Analysis of CSMA/CD and CSMA/CA

Apparatus Required:

Packet Tracer, End devices, switches,
connectors.

Requirements:

1. End device - They are the devices
through which we can pass the message
from one device to another
and they are interconnected.

2. Switch / Hub - Interface Between
two devices.

3. Cable - Used to connect the
devices.

Procedure:

Step 1: Click on end devices, select generic PC's drag and drop it on the window. Click on switch drag and drop it on the window.

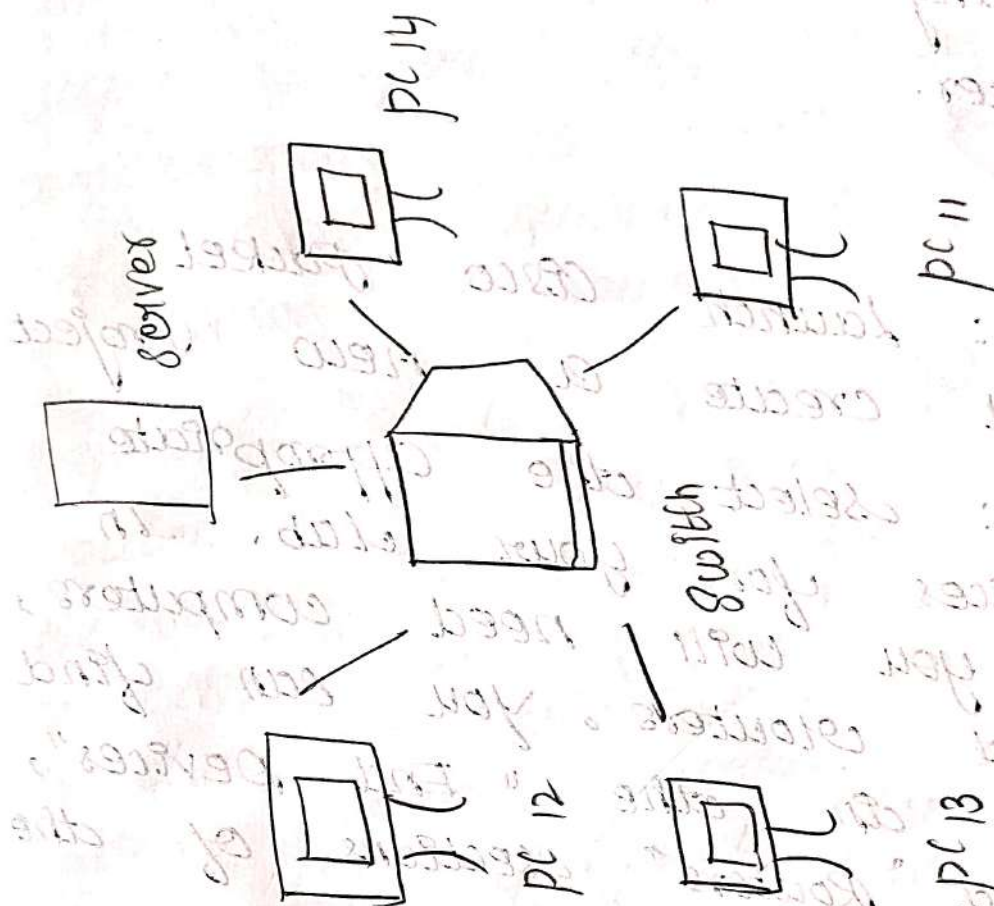
Step 2: Select the straight cable and connect all end device to switch. Assign the IP address for all end devices.

Step 3: Now set the IP address to host A (192.168.1.1) in static mode. Similarly set IP address for Host B (192.168.1.2) and Host C (192.168.1.3).

Step 4: To view the IP address give ip config command in command prompt. Using ping command, we can establish communication between two host devices.

Step 5: Now display the packet transmission in simulation mode.

DIAGRAM :



RESULT :

Thus the program executed successfully.

M. S. M.

Experiment - 10

Making computer Lab in Cisco Packet Tracer

Aim: Making computer Lab in Cisco
Packet Tracer.

Procedure:

Step 1: Launch Cisco Packet
Tracer and create a new Project.

Step 2: select the appropriate
network devices for your lab. In
this case, you will need computers,
switches and routers. You can find
these devices in the "End Devices",
"Switches" and "Routers" sections of the
device list.

Step 3: Drag and drop a switch
onto the workspace area. Connect the
switch to the power source by
clicking on the "Connection" option
and selecting "Power".

Step 4: Connect computers to the switch by dragging and dropping them onto the workspace area. click on the "connection" option and select "Fast Ethernet" to connect the computers to the switch.

Step 5: Repeat step 4 to add more computers to the lab. you can adjust the number of computers as per your requirements.

Step 6: Connect the switch to a router. Drag and drop a router onto the workspace area and connect it to the switch using a serial cable. To do this, click on the "Connection" option, select "Serial" and then select the appropriate serial Interface on the router.

Step 7: Configure IP addresses on the computers. select a computer, click on the Desktop tab in the device configuration panel, and configure the IP address, subnet mask, and default gateway for each computer.

Step 8: Configure IP addresses on the router interfaces. Select the router, click on the 'CLI' tab in the device configuration panel, and enter the interface configuration mode. Assign IP addresses to the router interfaces connected to the switch and computers.

Step 9: Test connectivity. Open the command prompt on each computer and try to ping other computers and the router's interfaces to ensure connectivity.

Step 10: Customize and expand the lab as desired. You can add additional devices, configure VLAN's, implement security measures, or set up servers within the Lab environment.

RESULT:

Thus the program has
successfully

Experiment - 11

Configuration of a simple static Routing in Packet Tracer Using a simple topology with two routers

AIM: To configure a router using packet tracer software and hence to transmit data between the devices in real time mode and simulation mode.

Apparatus Required:

Packet Tracer / End devices,
Hubs, connectors.

Procedure:

Steps for Building Topology:

- Step 1: Start Packet Tracer
- Step 2: choosing devices and connections
- Step 3: single click on the end devices
- single click on the generic host.

Place p0, p1 on topology area
connect p0 to switch 1.
Similarly place p2, p3 on
topology area for receiver side.

Router and place the router
between two switches. Connect
these switches into router through
connecting wires.

Step 4: Verifying Connectivity in
Real time mode

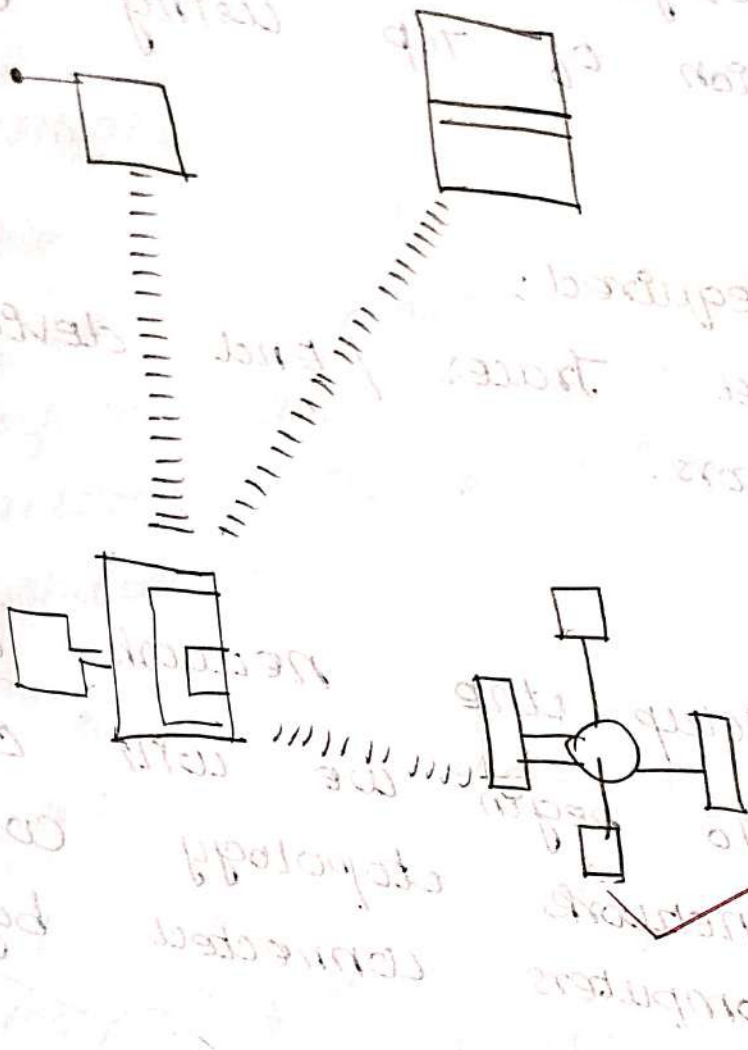
Be sure you are in Real
time mode.

Select the add simple pdu
tool used to ping devices.
Click once on p0, then once on

p3.
The pdu last status should show
as successful.

Step 5: Verifying connectivity in
simulation mode. Be sure you are
in simulation mode.

DIAGRAM.



RESULT:

Thus the program has been executed successfully.

Experiment - 12

Design the functionalities and Exploration of TCP using Packet Tracer

AIM: To design the functionalities and exploration of TCP using packet tracers.

Apparatus Required:
Packet Tracer / End devices,
Hubs, connectors.

Procedure:

Step: 1 Setup the network topology

Step: 2 To begin we will create a simple network topology consisting of two computers connected by a router.

Step 3: Configure IP addresses

Next, we will configure IP addresses for the computers. Double click on each PC to open the configuration

window and navigate to the desktop.

step 3: Configure the router

Now, we will configure the router. Double click on the router to open the configuration window and navigate to the cli tab.

COMMANDS:

enable

configure terminal

interface FastEthernet 0/0

ip address 192.168.1.254. 255. 255. 255.0

no shutdown

exit

step 4: Test the connection

Now that the network is set up and configured, we can test the connection between the two computers.

step 5: Explore TCP functionalities

Now let's explore the functionalities of TCP. We will use the Netcat

utility to establish a TCP
connection between the two
computers.

RESULT:

Thus the program has
been executed successfully.

Experiment - 13

Design the network model for Subnetting -
Class C Addressing Using Packet
Tracer

AIM:

To design the network model for
subnetting - class C addressing using
packet tracer.

Apparatus Required:

Packet Tracer / End devices, Hubs,
connectors.

Algorithm:

1. Determine the network requirements:
Identify the number of subnets and
hosts required for each subnet.
2. choose a subnet Mask: Select a
subnet mask that can accommodate
the required number of subnets and
hosts.
3. Calculate the subnet mask and
prefix length: Use the formula $2^p - 2 =$

where p is the number of host bits and n is the required number of hosts per subnet, to calculate the number of host bits required.

- 4: Configure the router: Configure the router interface with each other and with devices the hosts.
5. Test the network: Verify that the hosts can communicate with each other and with devices on other subnets.

Procedure :

Step: 1 click on end devices, select generic pc's drag and drop it on the window.

Step: 2 select the straight through cable and connect all end device to switch. Assign the IP address for all end devices.

Step: 3 To view the address, give

IP config command in command prompt.

Step 4: To view the IP address, give ipconfig command prompt.

Step 5: Now display the packet transmission in simulation mode.

RESULT:

Thus the program has been executed successfully.

22/11

Experiment - 14

Simulating X, Y, Z Company Network Design and simulate using Packet Tracer.

AIM:

To simulate X, Y, Z company network design and simulate using packet tracer.

Apparatus required:

Packet Tracer / End devices, Hubs, connectors.

Algorithm:

1. Identify the network requirements. Determine the number of users, devices and servers that will be connected to the network.

2. Create a network diagram: Use a network diagramming tool to create a visual representation of the network design, including the devices, servers, switches, routers and connections.

3. Configure the routers: Configure the routers with IP addresses, subnet masks and routing protocols as needed.

4. Configure the switches: Configure the switches with VLAN's and assign ports to each.

Procedure:

1. Start Packet Tracer: Launch Packet Tracer on your computer.

2. Create a new project: click on file and select New then select Network from the option.

3. Add devices: click on the devices tab in the bottom left corner of the window and drag and drop devices on the workspace.

RESULT:

Thus the program was executed successfully.

Experiment - 15

Configuration of DHCP (Dynamic host configuration protocol in Packet tracer)

AIM:
To configure DHCP (dynamic host configuration protocol) in packet tracer;

Apparatus required:

Packet tracer / End devices,
Hubs, connectors.

Algorithm:

1. Start
* set up the network topology in packet tracer with a DHCP server and DHCP clients connected to a switch.
2. Configure the DHCP server
* Assign an IP address to the server interface.

* Enable the DHCP service on the server.

* Define the IP address pool range that the server can assign to clients.

* Enable switch interfaces that connects to the DHCP clients.

3. Configure the DHCP clients: configure the clients to obtain their IP addresses automatically using DHCP, verify that the clients are set to use DHCP as the preferred method for IP assignment.

4. Client request and server response:

When a DHCP client boots up or its lease expires it sends a DHCP discover message as a broadcast on the local network.

5. Server Acknowledgement :

The DHCP server receives the request message and sends a DHCP acknowledgment (ACK) message to the client, confirming the IP address assignment and providing network.

Procedure :

1. Launch Cisco Packet Tracer and create a new network topology or open an existing one.
2. Add the necessary network devices to your topology, including a DHCP server, switch and DHCP clients. Connect them using appropriate cables.
3. Configure the DHCP server
 - * Select the DHCP server device and open its configuration panel.
 - * Assign an IP address to

the server interface connected to the switch.

* Enable the DHCP service on the server by checking the DHCP option.

4. Configure the switch:

* Select the switch device and open its configuration panel.

* Save the configuration.

5. Configure the DHCP clients:

* Select the each DHCP client device and open its configuration panel.

* Set the IP address assignment method to DHCP or obtain an IP address automatically.

RESULT:

Thus the program has been executed successfully.

Experiment - 16

Configuration of Firewall in Packet Tracer

Aim:
To configure firewall in
packet tracer.

Apparatus Required:

Packet tracer / End devices,
Hubs, connectors.

Procedure:

Step : 1 Set up the network topology
To begin we will create a simple
network topology consists of three
computers, a router and a firewall.

Step : 2 Configure IP addresses
Next we will configure IP addresses
for computers. Double click on each
pc to open the configuration window
and navigate to the Desktop tab.

Step: 3 Configure the Router.

Now, we will configure the router. Double click on the router to open the configuration window and navigate to the CLI tab. Enter the following commands:

Commands:

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.1.254. 255. 255.0
no shutdown
exit
```

Step: 4 Configure the firewall

Now we will configure the firewall. Double click on the firewall to open the configuration window.

Step 5: Test the connection

Now that the firewall is configured, we can test the communication between the computers. open a command prompt on pc1 and ping pc2 and pc3 by typing ping

Experiment - 17

Make a computer lab to transfer a message from one node to another to design and simulate using Cisco packet tracer.

Aim: To make a computer lab to transfer a message from one to another to design and simulate using Cisco Packet Tracer.

Apparatus Required:

Packet tracer / End devices, Hubs, connectors.

Procedure:

Step 1: Create the network topology
First, we need to create the network topology lab. In Packet tracer, drag two computers, a switch and two routers onto the workspace.

CODE:

PC1 PC2

11

Switch --- Router 1, --- Router 2

Step 2: Configure IP address

Next we will configure IP addresses for the computers. Double click on each PC to open the configuration window and navigate to the desktop tab.

Step 3: Configure the routers

Now, we will configure the routers. Double click on Router 1 to open the configuration window and navigate to the CLI tab.

Step 4: Configure routing

We need to configure routing between the routers so that they can communicate with each other. Enter the following commands on

Router 1.

Step 5: Send a message

To send a message from PC1 to PC2, open the command prompt on PC1 and type: ping 192.168.12

Experiment - 18

Simulation of a multi-hop network
on a packet switch

Algorithm:
1. To estimate the packet delay
in the network, we need to know
the packet size and the
connection.

Procedure:
Step 1: Form a new packet.
Step 2: Select the appropriate network
for your computer, select
you will need multimedia
and IP
as IP phones and
Step 3: Design the network
of connections.

RESULT:

Thus the Program has
been executed successfully.

Experiment - 18

Simulate a multimedia network
in cisco packet Tracer.

Aim:

To simulate a multimedia network in cisco packet Tracer.

Apparatus Required:

Packet Tracer / End devices, Hubs, connectors.

Procedure:

Step: 1 Launch cisco packet tracer and create a new project.

Step: 2 Select the appropriate network devices for your multimedia network. You will need computers, switches, routers and multimedia devices such as IP phones and IP cameras.

Step: 3 Design the network topology. Determine the layout of your network and the connections between devices.

Step: 4 Drag and drop the devices onto the workspace area. Connect the devices using appropriate cables or wireless connections.

Step: 5 Configure IP addresses on the devices. Assign IP addresses, subnet masks and default gateway to the computers, IP phones and IP cameras.

Step: 6 Set up multimedia services. Configure the necessary services for multimedia communication, such as VoIP (voice over IP) for IP phones and streaming protocols for IP cameras.

MAN

RESULT: Thus the program has been executed successfully.

Experiment - 19

IoT Based Smart Home Applications

Aim:

To implement IoT based smart home applications in Cisco Packet Tracer.

Apparatus Required:

Packet Tracer / End devices, Hubs, connectors

Procedure:

Steps

1. Create a network topology in Cisco Packet Tracer that includes IoT devices such as sensors, actuators and gateways.
2. Configure the IoT devices with appropriate IP addresses, subnet masks and gateway addresses.
3. Setup for a communication protocol between the IoT devices using MQTT, CoAP, or any other protocol of your choice.

4. Write a code to collect data from the sensors and send it to the gateway.

5. Use the gateway to process the data and sends commands to the actuators.

6. Finally we use a web interface or mobile application to monitor and control the IOT devices. This can be used for various applications such as home automation, smart cities and industrial automation.

Done

RESULT:

Thus the program has been executed successfully.

Experiment - 20

Implementation of IOT Based Smart Gardening.

Aim: To implement IOT Based smart gardening using Cisco Packet Tracer.

Apparatus Required:
Packet Tracer / End devices,
Hubs, connectors.

Procedure:

Step 1: Create a new project in Cisco Packet Tracer and drag a generic IOT device from the IOT devices section into the workspace.

Step 2: Right click on the IOT device and select config / Attributes

Step 3: In the configuration tab, select the device IOT server from the drop down list. You can choose Cisco IOT cloud or another cloud service of your choice.

Step : 4 In the attributes tab,
add the following attributes:

- * Temperature
- * Humidity
- * Soil moisture
- * Light Intensity.

Step : 5 Create a soil moisture sensor
and a light sensor from the sensors
section of the devices panel. Drag
and drop these sensors onto the
workspace.

Step : 6 connect the sensors to the IoT
device using the wiring tool.

25/11

RESULT:

Thus the program has been
executed successfully.

Experiment - 21

Implementation of IOT devices in Networking

Aim: To implement an IOT device in networking using Cisco Packet Tracer.

Apparatus Required:
Packet Tracer / End devices,
Hubs, connectors.

Procedure:

Steps:

1. Open Cisco Packet Tracer and create a new project. Drag and drop a router from the devices panel onto the workspace area.

2. Connect the router to the Internet by dragging and dropping a cloud device from the devices panel onto the workspace area and then connecting the router to the

cloud using a straight through cable.

3. Add an IOT device into the network by dragging and dropping a device from the devices panel onto the workspace area. These are various IOT devices available in the devices panel such as a Raspberry Pi or an Arduino.

4. Connect the IOT device to the router using an ethernet cable. To do this, check on the IOT device tab and then click on the config section under the ethernet interface and select the ethernet interface and then click on the '+' button to add a new interface to the router. Connect the new interface to the router.

5. Configure the IOT device by clicking on it and then clicking on the CLI tab. This will bring up the command line interface for the IOT device, where you can configure its settings.

6. Test the connectivity of the IoT device by pinging it from the router or from another device on the network.

RESULT:

Thus the Process has been executed successfully.

Experiment - 22

IoT based AAA Local and server based authentication configuration

AIM: Designing an IoT based AAA local and server based authentication configuration.

Apparatus required:

Packet Tracer / End devices, Hubs, connectors.

Procedure:

1. Define the components:
IoT devices: These are the devices that need to be authenticated and authorized to access the network resources.

Local AAA server: This server will handle authentication requests locally.
Central AAA server: This server will provide an additional layer of authentication and authorization.

for higher level access control.

2. Setup Local AAA Server:
Configure the local AAA server with the necessary software and databases to handle authentication and authorization requests.

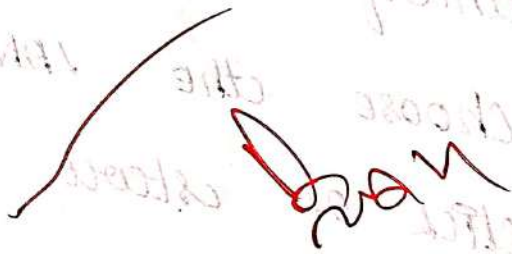
Define user profiles or roles and their associated permission on the local AAA server. Set up a secure communication channel between the IoT devices and the local AAA server.

3. Implement Local Authentication:
When an IoT device wants to connect to the network, it sends an authentication request to the local AAA server.

4. Implement Local Authorization:
Once authenticated, the IoT device sends an authorization request to the local AAA server. The local AAA server checks the user profile or role associated with

the IOT device and verifies if it has the necessary permission to access the requested resources. If authorized the local AAA server sends an authorization response to the IOT device.

5. Configure central AAA server:
Set up a central AAA server that will provide an additional layer of authentication and authorization for critical resources or higher level access control.



RESULT:

Thus the program has been executed successfully.

Experiment - 23

Transport Layer Protocol Header
analysis using wire shark
tcp and udp.

Aim: To analyze capturing of
Transport layer protocol header analysis
using wire shark tcp and udp.

Software used:

wire shark network analyzer.

Procedure:

1. Open wire shark.
2. Click on dist the available
capture interface.
3. Choose the LAN interface.
4. Click on start button.
5. Active packets will be displayed.
6. Capture the packets and
select any ip address from the source.

7. Click on the expression and select IPv4 - IP address source address in the field name

8. Select the double equals (=) from the selection and enter the selected IP source address.

9. Click on apply button.

10. All the packets will be filtered using source address.



RESULT :

Thus the program has been executed successfully.